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Comparison of fly ash and rice husk ash effects on the mechanical properties of roller compacted concrete for pavements

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ABSTRACT

Laboratory investigation was conducted focused on the utilisation of waste materials comprising Class F fly ash (FA) and rice husk ash (RHA), in roller compacted concrete (RCC) for pavement applications. The mixtures of RCC comprised CEM I 52.5 N Portland cement containing 0%, 5%, 10%, 20% and 30% wt FA or RHA. Various aggregate characteristics were measured to determine their suitability for road applications. Compaction tests were done to obtain values of optimum moisture content for RCC mixtures. The mechanical properties investigated comprised compressive strength, static modulus of elasticity, dynamic modulus of elasticity and ultrasonic pulse velocity. Results showed that proportions of up to 5–10% wt RHA or FA were optimal and can be recommended for incorporation in RCC pavement mixtures. Standard models giving the relationships between compressive strength, static and dynamic elastic moduli of normal concretes, were found to be also applicable to RCCs.

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Roller compacted concrete; pavements; compressive and tensile strength; static and dynamic elastic modulus; ultrasonic pulse velocity

1. Introduction

Over the years, roller compacted concrete (RCC) has gained wide acceptance as a viable construction material used in mass concrete applications, especially dam construction and roadworks. RCC is generally similar to normal or conventional concrete except for its zero – slump mixture typically prepared at higher fine aggregate content, along with lower cement and water contents. Accordingly, RCC is a stiff mixture suitable for pavement construction, wherein compaction is done using the asphalt roller compactor, hence the use of the name ‘roller compacted’. Typically, RCC is capable of carrying light vehicular traffic immediately after pavement placement. Unlike conventional concrete roads, RCC pavements do not involve the embedment of steel reinforcement; however, they are generally dense, durable and strong with high load-bearing capacity (Hafez et al., 2023; Harrington et al., 2010). The total aggregate content of RCC pavement mixtures is typically higher, being in the range of 75–85% by volume compared to 60–75% by volume for normal concrete pavements (ACI, 1995; Harrington et al., 2010). Construction of RCC pavements does not involve the installation of joints, formwork, dowels or steel reinforcement, which simplifies practical works, making the application fast and economical (Harrington et al., 2010).

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RCC was reportedly first used in the 1930s for the construction of low-traffic roads in Sweden. In the 1940s and 1980s, RCC was used for the construction of airport runways and military facilities in the USA, respectively. This pavement technology has presently gained worldwide acceptance for use in highway engineering and heavy-duty and light industrial applications (FHWA, 2016; Harrington et al., 2010). Accordingly, RCC has been used to construct parking or storage areas for heavily loaded trucks and aircraft, and to construct runways in airports, highway shoulders and local streets, amongst others (ACPA, 2014). Such RCC applications have been implemented in various countries including the USA, UK and South Africa, amongst others (CSRA, 1987; Du Plessis et al., 2014; FHWA, 2016; UK Government, 2021).

This study focused on the utilisation of fly ash (FA) and rice husk ash (RHA) in RCC. Both materials are waste by-products and their utilisation in concrete has several engineering and environmental benefits, along with cost reduction. These pozzolanic materials react with the calcium hydroxide product arising from cement hydration, to form secondary calcium – silicate – hydrate (C-S-H), thereby reducing permeability and improving durability, generally. Their pore-infilling effect also promotes the formation of denser microstructure, thereby enhancing mechanical properties (Fulton's, 2001; Neville, 1996).

It may be recalled that pozzolanic materials, including FA, RHA, ground granulated blast – furnace slag (GGBS), silica fume (SF) and others, are already specified in cement standards (EN 197-1, 2000) for incorporation as supplementary cementitious materials (SCMs). It is also well – established that the production of Portland cement is responsible for 5–8% CO₂ emissions into the atmosphere. As such, incorporation of SCMs into concrete mixtures contributes to the environmental benefit of reducing CO₂ emissions. Besides, SCMs are known to generally improve the durability properties of concretes (Fulton's, 2001; Naghizadeh & Ekolu, 2017; Neville, 1996).

1.1. Roller compacted concrete for pavements

1.1.1. Specifications

The UK national standards for highways, for example, have developed a design chart for estimating the concrete thickness layer of RCC pavements (National Highways UK, 2021). For quality control, the minimum compressive strength of 50 or 5 MPa minimum flexural strength, is specified as the basis for layer thickness design (UK Government, 2021). According to CSRA specification for concrete roads, a minimum 28-day compressive strength of 38 MPa or corresponding flexural strength of 3.8 MPa, is considered adequate to provide sufficient load-bearing capacity and abrasion resistance (CSRA, 1987). Generally, the 28-day compressive strength of 28–41 MPa, along with flexural strength values of 3.5–7 MPa are typically used or specified. It is suggested in the literature that high-performing RCC pavements should typically attain compressive and flexural strengths of 40 and 4.5 MPa, respectively (Rambabu et al., 2023; Selvam et al., 2022). RCC pavement construction works can be fully opened to normal traffic, upon attaining a minimum compressive strength of 13.8–17.2 MPa (Harrington et al., 2010).

1.1.2. Mixture design

RCC mixtures differ widely from those of conventional concrete. The sand content of normal concrete mixtures is usually 35–50%, being typically lower than stone content. In contrast, RCC mixtures usually have higher sand content of about 50–55%, while stone content can be below 40%. As mentioned earlier, the binder content of RCC mixtures is also lower than that of normal concretes. The minimum cementitious content for RCC pavements is typically 250 kg/m³ or 10%, whereas that of normal concretes is higher, being typically in the range of 10–15% (FHWA, 2016; Harrington et al., 2010; Rambabu et al., 2023). Interestingly, studies have also been done to investigate RCC mixtures of very low 80–160 kg/m³ cementitious content (Villena et al., 2011). Another crucial mix design aspect that distinguishes RCC from normal concrete, is water content. For normal concrete, water content is a major variable that is adjusted in mix design along with cement content, to obtain the required workability

and concrete strength. In contrast, the amount of water used in RCC mixtures is not prescribed in a similar manner as it is done for normal concrete. Rather, the water content used in RCC mixtures is a fixed quantity determined as the optimum moisture content (OMC) based on the moisture–density relationship obtained using the soil–compaction approach (ASTM D1557, 2012). During construction, it is desirable for the pavement to develop enhanced early–age strength or load-bearing capacity, which subsequently allows early opening of the road to traffic. Accordingly, RCC mixtures are typically made with continuously – graded combined aggregate, which promotes interlocking of aggregate particles of different size fractions (FHWA, 2016). Being zero – slump mixtures, the workability of RCCs is more fittingly considered as consistency and is measured using the Vebe test method (ASTM C1170, 2008). Otherwise, most other mechanical properties of RCC including compressive strength, splitting tensile strength, modulus of elasticity and others, are measured in a manner similar to those of normal concrete.

1.2. Roller-compacted concrete containing fly ash or rice husk ash

1.2.1. Influence of fly ash

Most desirable engineering properties of RCC pavements are similar to those of conventional concrete pavements including, fresh concrete properties, mechanical properties such as compressive strength, splitting tensile strength, flexural strength, modulus of elasticity, abrasion resistance, fatigue behaviour, bond strength and durability – related properties including porosity, water absorption, alkali-silica reaction, amongst others (ACI, 1995; Hafez et al., 2023; Selvam et al., 2022; Tayabji & Okamoto, 1987). It is known from various literatures that incorporation of FA into mixtures positively impacts RCC properties (Hafez et al., 2023; Mohamad et al., 2024; Solikin et al., 2025).

Rao et al. (2015; 2016) incorporated proportions of 0%, 20%, 40% and 60% FA into RCC mixtures made at 295 kg/m³ cementitious content. They reported that OMC increased with a rise in FA content, from 5.7% for the control to 7.1% for the mixture containing 60% FA, while compressive, flexural and splitting tensile strength values decreased correspondingly.

In the investigation conducted by Mardani-Aghabaglou and Ramyar (2013), proportions comprising 0%, 20%, 40% and 60% FA were incorporated as partial replacements of cement or fine aggregate. The cementitious content of 250 kg/m³ was used in all mixtures. They reported that partial replacement of cement with FA led to a decrease in compressive, flexural and splitting tensile strength results. In contrast, the opposite effect was observed for RCC wherein fine aggregate was replaced with FA as these mixtures exhibited increases in mechanical properties with a rise in the content of the SCM. Their findings are consistent with those reported by Seav et al. (2023) showing that incorporation of 0%, 25%, 50% and 100% bottom ash as sand replacement, led to an increase in compressive and flexural strength values of the mixtures. Cao et al. (2000) also reported that while RCC mixtures containing up to 55% FA as sand replacement initially showed lower strength, the results exhibited greater late strength gain which at 90 days exceeded that of the control.

1.2.2. Influence of rice husk ash

Rice is an important and popular food staple worldwide, more so in populous developing countries including China, India and Indonesia, amongst others. As such, rice production volume worldwide is exorbitant, having recently reached about 787 million tonnes per year (FAO, 2022). Each tonne of rice produces 20% raw husk and 4% RHA as agricultural waste material (Mehta, 1992; Villena et al., 2011). However, very limited research presently exists on the potential use of RHA in RCC. Moreover, it is necessary to distinguish between the use of RHA for partial cement replacement versus its use as sand replacement material. Research studies show that incorporation of RHA as fine aggregate in RCC typically increases compressive strength (Meghwar et al., 2024). Such a study by Villena et al. (2011) found 5–7% RHA to be the optimum proportion for the use of the SCM as sand replacement material in RCC. Vahidi et al. (2017) reported that the use of 9% RHA as cement replacement material gave optimum compressive strength, but results decreased at SCM proportions above 15%

RHA (Aghaeipour & Madhkan, 2020). Similar observations, as discussed in the foregoing, have been reported regarding the use of RHA as cement replacement material in normal concrete, wherein proportions of 7.5–10% RHA reportedly gave optimum strength and durability performance, depending on cementitious content (Ahmed et al., 2020; Ahmed & Kamau, 2017; Kamau et al., 2018).

In an investigation by Modarres and Hosseini (2014), proportions of 3–5% RHA were incorporated as cement replacement material in RCC. Their results showed an increase in OMC and a reduction of MDD owing to the use of RHA. Panpipat and Cheerarot (2022) incorporated 0%, 10%, 20% and 30% RHA as cement replacement material in RCC mixtures made at 10%, 12% and 14% binder content. They reported that RCC mixtures containing up to 10% RHA gave similar compressive strength values as the control, but at higher proportions of the SCM, strength decreased correspondingly. Ahmed et al. (2020) reported similar observations, as in the foregoing, showing that RCC mixtures containing up to 10% wt RHA gave pavement concrete strength that was similar to or higher than that of the control mixture.

1.3. Objectives and research significance

The objectives of the present study were to investigate the performance responses of RCC mixtures containing FA which is a by-product from coal-power stations, along with those containing RHA which is an agricultural waste material. In the present investigation, RCC mixtures were prepared using CEM I 52.5N Portland cement containing 0–30% wt FA or RHA. Various material characteristics of aggregates were measured including relative density, water absorption, particle size distribution, Aggregate Crushing Value (ACV) and Los Angeles Abrasion (LAA). The mechanical properties of RCC mixtures measured comprised:- compressive strength development, splitting tensile strength, ultrasonic pulse velocity (UPV) and static and dynamic moduli of elasticity. As an additional objective, the standard relationship models for the foregoing mechanical properties of normal concretes were evaluated for potential applicability to RCCs.

The practical significance of this research is related to the engineering and environmental benefits of utilising RHA and FA in RCC mixtures, along with a reduction in the cost of pavement construction. Presently, RCC pavement construction is hardly or not at all employed in most developing countries. It is, however, well – established that RCC is an economical material system that allows fast pavement construction and could contribute beneficially to engineering applications. It is also essential to consider that bituminous mixtures are presently the dominant pavement construction material in most developing countries. As such, employment of RCC for pavement construction not only reduces fossil fuel consumption since bitumen is a by-product of petroleum, but it would also increase the utilisation of waste materials such as FA and RHA, thereby promoting sustainability.

2. Experimental programme

2.1. Materials

The cementitious materials used to prepare RCC mixtures were CEM I 52.5 N Portland cement supplied by Pretoria Portland Cement (PPC) (Pty) Ltd, along with FA obtained from Eskom SOC Ltd. The RHA used was supplied by E-coco Products UK as part of an ongoing international collaboration with Leeds Beckett University. Table 1 shows the chemical compositions of the various cementitious materials, as determined using X-ray fluorescence analysis. It can be seen that RHA contained predominantly SiO_2 , hence its composition was similar to that of SF, while FA was clearly an aluminosilicate material. It can thus be expected that the two SCMs comprising FA and RHA, would exhibit different effects on RCC performance.

The aggregate grading requirements for RCC pavement construction are different from those of its use for dam construction. For RCC pavement construction, the crucial aggregate grading requirement specified by the US Army Corps of Engineers (Fulton's, 2001; USCORPS, 1988) is 18–36% passing the

Table 1. Chemical compositions of the cementitious materials used.

Composition		Portland cementCEM I 52.5 N	Fly ash(FA)	Rice Husk Ash(RHA)
Oxides and ignition loss (%)	SiO ₂	20.57	56.45	89.20
	Al ₂ O ₃	4.49	30.27	0.37
	Fe ₂ O ₃	3.07	3.58	0.60
	MgO	1.11	1.06	1.53
	CaO	64.47	4.59	0.73
	TiO ₂	0.53	1.57	0.03
	P ₂ O ₅	0.08	0.38	1.08
	Na ₂ O	0.05	0.14	0.10
	K ₂ O	0.32	0.77	1.71
	LOI ^a	4.04	0.42	3.88
Density	Relative density	3.14	2.24	1.12

^aLOI: Loss on ignition.

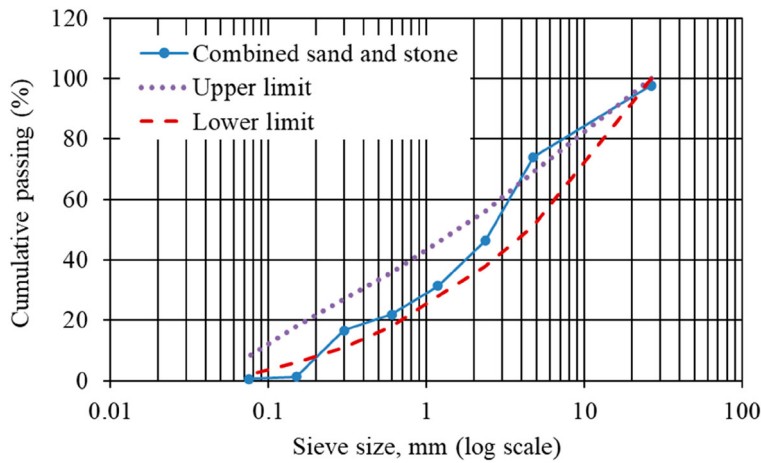


Figure 1. Grading of aggregate (Fulton's, 2001; USCORPS, 1988).

0.60 mm sieve, which is consistent with that of continuously graded aggregate to promote interlocking, compared to the higher percentages of 40–60% for gap graded aggregate of normal concrete. The aggregate used in the present investigation was 19 mm quartzite stone and quartzite crusher sand, giving the combined grading shown in Figure 1. Also plotted in the figure are the upper and lower grading limits specified by the US Corps of Engineers (Fulton's, 2001; USCORPS, 1988).

The aggregate employed in the present study was also evaluated to determine its suitability for use in pavement construction. Accordingly, material characteristics of the aggregate were measured including relative density, water absorption, ACV and LAA. The density and water absorption values of stone were tested in accordance with SANS 3001 – AG20 (2014), while those for sand were measured as per AASHTO T84 (2013). The ACV and LAA tests were done in accordance with SANS 5841 (2008) and SANS 5846 (2006), respectively. Table 2 gives the measured physical properties of the aggregate. It can be seen that both the sand and stone, were good quality aggregate meeting the various specified requirements. For example, the low values of ACV = 17.0% and LAA = 23.8% measured for stone, are within the respective maximum limits of 29% and 25%, generally recognised for the use of aggregate in road construction (Stålheim, 2014; Tan et al., 2022).

2.2. Mixtures and methods

The SCMs used were incorporated into CEM I 52.5N cement at proportions of 0%, 10%, 20% and 30% wt FA or RHA. These proportions of the SCMs were chosen considering findings from the various literature

Table 2. Physical properties of aggregate.

Aggregate	Property	Measured value	Limit for use in wearing course (%)
Sand	Relative density (RD)	2.58	
	Water absorption (WA), %	1.09	
Stone	Relative density (RD)	2.65	
	Water absorption (WA), %	0.85	2
	Aggregate Crushing Value (ACV), %	17.0	29
	Los Angeles Abrasion (LAA), %	23.8	25

Table 3. Mixtures of roller-compacted concrete containing fly ash (FA) or rice husk ash (RHA).

Sample ID	Proportions		Quantities of ingredients (kg/m ³)					
	FA (%)	RHA (%)	CEM I 52.5N	FA	RHA	Water	19 mm stone	Crusher sand
OFA	0	–	201	0	0	152	960	1100
5FA	5	–	191	10		152	960	1100
10FA	10	–	181	20		152	960	1100
20 FA	20	–	161	40		152	960	1100
30 FA	30		141	60		152	960	1100
5RHA	–	5	191		10	152	960	1100
10RHA	–	10	181		20	152	960	1100
20RHA	–	20	161		40	152	960	1100
30RHA	–	30			60	152	960	1100

discussed in Section 1.2. The cementitious content of RCC mixtures was 201 kg/m³ or 8.6%, while water content of each mixture was determined using the soil-based compaction approach employed in the standard methods (ASTM D1557, 2012; BS, 1377-4, 1990; SANS, 3001-GR30, 2015). The soil compaction method is the most widely adopted mixture proportioning approach for RCC pavements. The other proportioning methods, comprising the Concrete Consistency Method, Solid Suspension Model and Optimal Paste Volume Method, are typically employed to design RCC mixtures for non-pavement applications such as dam construction. To carry out the compaction test, the water content added to each RCC mixture was varied to 4.5%, 5.5%, 6.5%, 7.5% and 8.5% of the total mass. The water content interval of 1.0%, as well as the lower and uppermost values employed, were determined by trial and error based on the behaviours of the RCC mixtures. During casting, cylindrical moulds of size 142.4 dia. x 116.4 mm height were each filled in three layers, each layer being subjected to 55 blows using the 4.5 kg proctor hammer. The results obtained were plotted to determine the compaction curve giving the moisture–density relationship of each mixture. The OMC results obtained from compaction curves were subsequently used as water content values of the RCC mixtures.

Table 3 gives the various RCC mixtures that were prepared. A 50 L concrete pan mixer was used for mixing, into which dry materials were added and then mixed for one minute. With the mixer running, water was added slowly over one minute, and then mixing was continued for further two minutes. Nineteen (19) cubes of 100 mm size and three (3) large cylinder samples, were cast per mixture.

The samples were cast and then cured in water at room temperature of 23°C, until testing at various ages (ASTM C192, 2019; SANS, 5861-3, 2006; SANS, 5861-1 2006). Compressive strength was measured at the various ages of 3, 7, 28 and 120 days, as per SANS 5863 (2006) and BS 1881–116 (1983). The 28-day splitting tensile strength was also measured by testing 100 mm cube samples (GB/T, 50081, 2019). Also cast were the large 150 dia x 300 mm cylinders used to measure ultrasonic pulse velocity (UPV) (ASTM C597, 2022) and static modulus of elasticity (E_c) (ASTM C469, 2002). Figure 2 shows the various experimental test set-ups. Equation (1) was used to calculate the dynamic modulus of elasticity (E_d) from UPV results. The various foregoing tests are standard methods which collectively provide a comprehensive evaluation of mechanical properties. The experimental programme conducted is

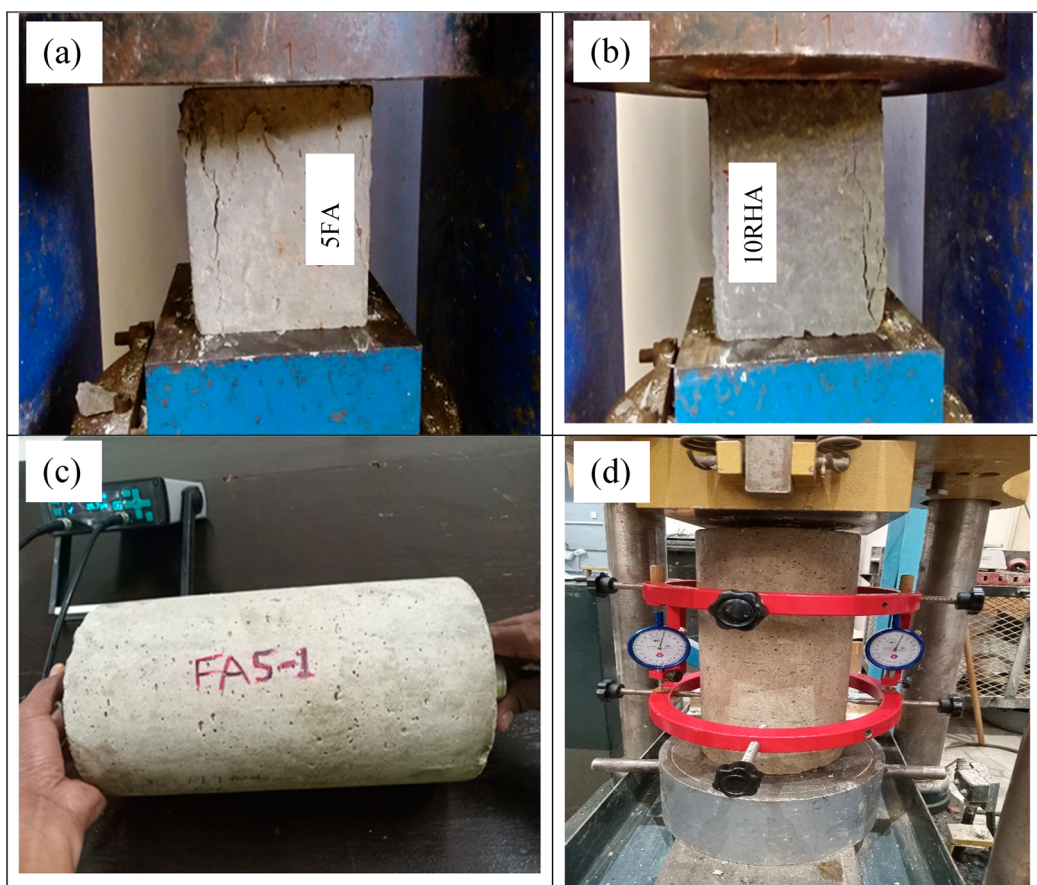


Figure 2. Experimental set-ups showing testing of (a,b) compressive strength, (c) UPV and (d) static elastic modulus.

shown in the flow chart given in Figure 3, which also includes the different standard test methods used.

$$E_d = \frac{\rho(\text{UPV})^2(1 + \mu)(1 - 2\mu)}{1 - \mu} \quad (1)$$

where E_d is the dynamic modulus of elasticity, ρ is the density of concrete (kN/m^3), UPV is the ultrasonic pulse velocity (km/s), μ is poison's ratio assumed to be 0.2.

3. Results and discussion

3.1. Optimum moisture content results

Figure 4(a,b) shows the compaction curves of RCC mixtures, giving the variation in dry density with a rise in moisture content from 4.5% to 8.5%, under the influence of different FA or RHA proportions. The OMC of the control mixture was 6.5%. It can be seen in Figure 4(a) that although the curve profiles were different, the OMC value for MDD was consistently 6.5% regardless of FA proportion, implying that the SCM had no significant influence on the water content of the RCC mixtures. However, the influence of RHA was somehow different from that of FA. Figure 4(b) shows that at lower proportions of up to 10% wt RHA, the SCM had no significant effect on OMC, instead it maintained 6.5% water content. But at 20% and 30% wt RHA, the OMC shifted to the higher water content of 7.5%, implying that the use of higher proportions of the SCM led to an increase in water demand of the RCC mixtures. Similar findings

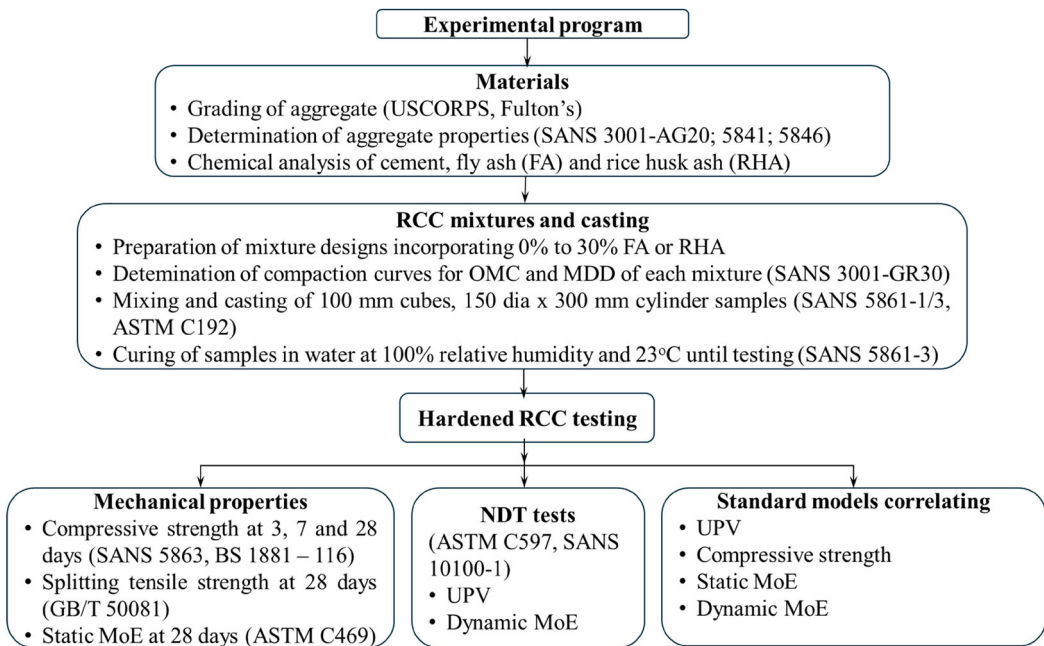


Figure 3. Experimental programme: FA – fly ash, RHA – rice husk ash, OMC – optimum moisture content, MDD – maximum dry density, MoE – static modulus of elasticity and UPV – ultrasonic pulse velocity.

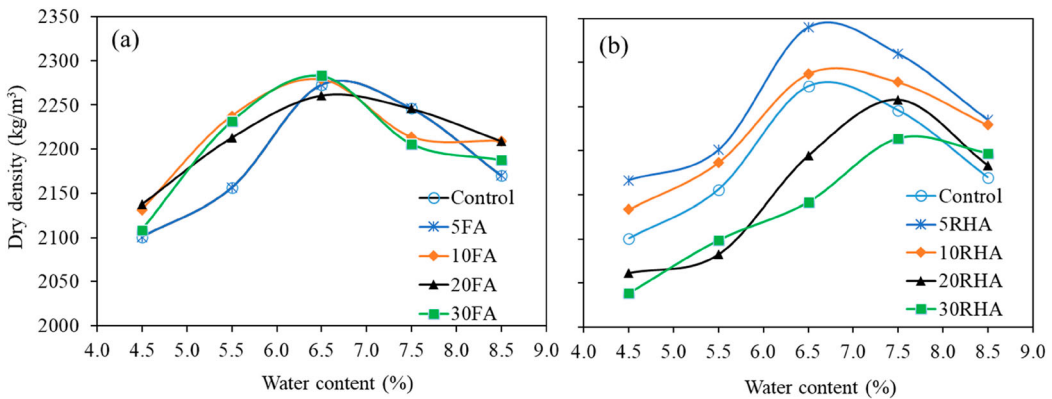


Figure 4. Moisture–dry density relationships as influenced by (a) fly ash (FA) and (b) rice husk ash (RHA).

were reported by Jain et al. (2021), wherein RHA was incorporated in normal concrete. It may be noted that the density of RHA is much lower than that of cement or FA (Table 1), hence mixtures containing this SCM had a higher paste volume and higher water absorption, thereby reducing workability. Also at higher RHA proportions, less water is available for cement hydration, which increases the coarseness of the microstructure. However, RHA's pore-filling effect can be an overriding factor leading to the formation of dense microstructure, reduced porosity or permeability and overall improved durability (Modarres & Hosseini, 2014).

3.2. Compressive strength results

Shown in Figure 5(a,b) are the strength effects of FA and RHA on RCC. In both cases, compressive strength generally decreased with a rise in the proportion of the SCM incorporated, as expected.

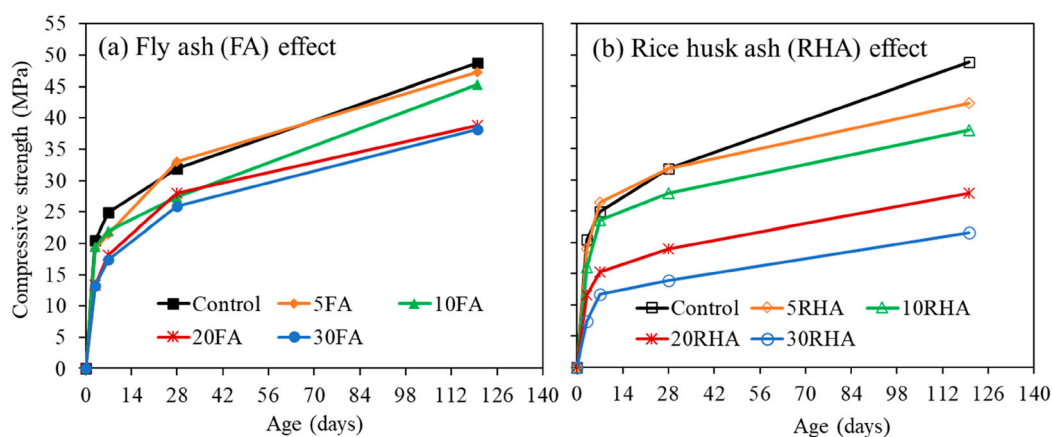


Figure 5. Compressive strength of RCC mixtures containing (a) fly ash (FA) and (b) rice husk ash (RHA).

However, significant differences can be seen between the effects of FA and those of RHA. It is evident that the 5–10% wt FA mixtures showed up to 13.6% reduction of early age 7 – day compressive strength, beyond which the mixtures showed late strength gains that caught up to give similar long-term values as those of the control. In contrast, the 5–10% wt RHA mixtures significantly enhanced early-age 7 – day compressive strength exhibiting up to a 6.4% increase, but gave reduced long-term strength relative to that of the control. The tendency exhibited by FA is typical of pozzolans that possess significant quantities of both silica and alumina, being usually within the range of 12–40% SiO_2 and 30–60% Al_2O_3 . Such SCMs comprise natural pozzolans including volcanic ash, metakaolin, zeolite or artificial pozzolans such as FA, GGBS and others (Ekolu et al., 2006; Naghizadeh & Ekolu, 2017; Sinngu et al., 2022; 2023).

Meanwhile, silica-rich SCMs tend to be more effective in enhancing early strength but do not exhibit a significant effect on long-term strength growth. Among conventional SCMs of this category is SF, which is known for enhancing early strength, especially for high-strength concretes. SF typically contains $> 90\%$ SiO_2 content, and is usually incorporated into concrete at low proportions of about 5–15% wt (Fulton's, 2001; Nochaiya et al., 2010). Similarly, the RHA used in the present study was a silica-rich pozzolan containing a high 89.2% SiO_2 content (Table 1), which underscores its early – strength-enhancing effect on RCC, as seen for the 5–10% wt RHA mixtures (Figure 5(b)).

Figure 6 compares the compressive strengths of RCC mixtures containing FA versus those containing RHA. It can be seen that the strength values of 5–10% wt RHA mixtures, were generally similar to those of corresponding FA mixtures. But upon a rise in SCM proportions to higher levels of 20–30% wt, the performance of RHA mixtures significantly diminished to levels lower than those of the corresponding FA mixtures. Indeed, strength values of the 20% and 30% wt RHA mixtures were respectively 13–30% and 44–50% lower than those of the corresponding FA mixtures. This observation can be partly attributed to the lower relative density of RHA. Thus at higher SCM proportions, the paste volume of mixtures containing RHA was much higher than that of the control. Furthermore at higher SCM content, once the calcium hydroxide arising from cement hydration has been depleted due to pozzolanic reaction, the unreacted pozzolanic material has minimal (sometimes adverse) contribution to strength, merely fulfilling the role of a filler (Ahmed et al., 2020).

It is evident from the foregoing observations that, the potential use of RHA in RCC mixtures is most effective at proportions not exceeding 10% wt. This finding is consistent with those in the various literature reporting 5–10% wt RHA to be the optimum proportions for the use of the SCM in normal concretes (Amran et al., 2021; Bhowmik & Pal, 2023; Fitriani et al., 2022). Evidently, RCC mixtures containing 10% FA and 10% RHA gave 28-day compressive strengths of 27.3 and 28 MPa respectively,

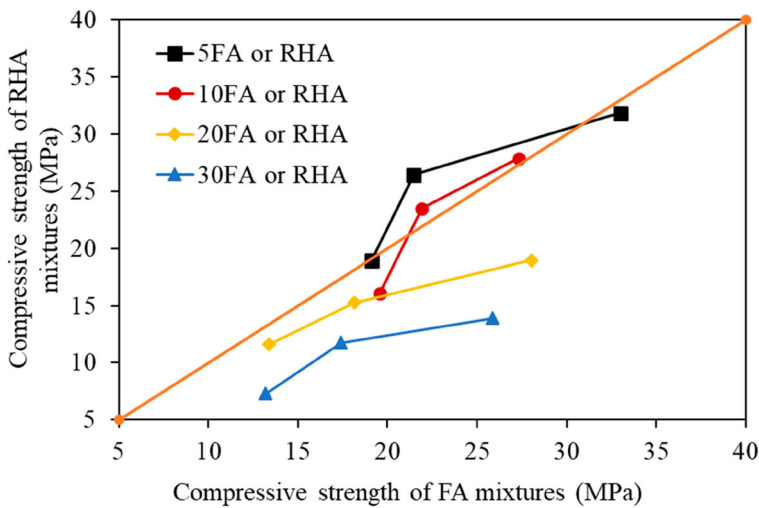


Figure 6. Comparison of 3-day and 28-day compressive strengths of RCC mixtures containing fly ash (FA) versus those containing rice husk ash (RHA).

which generally meet the required 28–41 MPa typical values specified for RCC pavement mixtures (Section 1.1.1).

3.3. Splitting tensile strength results

Figure 7 shows the 28-day splitting tensile strength results of RCC mixtures containing FA and RHA. Evidently, the tensile strength values of both pozzolans reduced with a rise in SCM content, as similarly seen for compressive strength results (Figure 5). Interestingly, the reductions in tensile strength results of FA mixtures were consistently within 16–23% regardless of the SCM proportion. In contrast, RHA mixtures exhibited a steady decline in tensile strength values with a rise in SCM proportion from 0% to 30% wt. A drastic drop in tensile strength results occurred at the higher proportions of 20–30% wt RHA, giving reductions of 50–60%. These observations indicate the absence of significant late strength gain in RCC mixtures containing RHA, whereas the mixtures containing FA exhibited long-term strength growth attributed to pozzolanic reaction, thereby attaining significant tensile strength values even at higher proportions of the SCM (Figure 7). It is well – established that SCMs densify the interfacial transition zone (ITZ), a relatively porous and weak microstructural aggregate–paste interface typically prone to crack initiation during concrete failure (Nduka et al., 2022). The splitting tensile strength values of mixtures containing up to 10% wt FA or 10% wt RHA, were all between 2.0 and 2.5 MPa. But at higher proportions above 10% wt, the FA mixtures maintained similar splitting tensile strength values averaging about 2.3 MPa, while the corresponding values for RHA mixtures reduced markedly to about 1.3 MPa. The foregoing contrasting observations can be attributed to different effects of FA and RHA on ITZ densification, owing to C-S-H formation in mixtures containing the SCMs (Meenakshi et al., 2019; Nduka et al., 2022; Sadrmomtazi et al., 2018). Expectedly, the localised effect of ITZ densification due to the pozzolanic reaction, enhances tensile strength more significantly while showing little to no impact on compressive strength. The excessive paste volume of mixtures containing higher proportions comprising 20–30% wt RHA, may be responsible for the observed drastic reduction in tensile strengths of the mixtures.

As typical of concretes generally, a correlation between compressive strength and tensile strength values of the RCC mixtures can be seen in Figure 8. While a direct and strong correlation is evident for RHA mixtures, the trend for FA mixtures seemed distorted most likely due to the ITZ densification effect that does not enhance compressive strength to the same extent at which it increases tensile

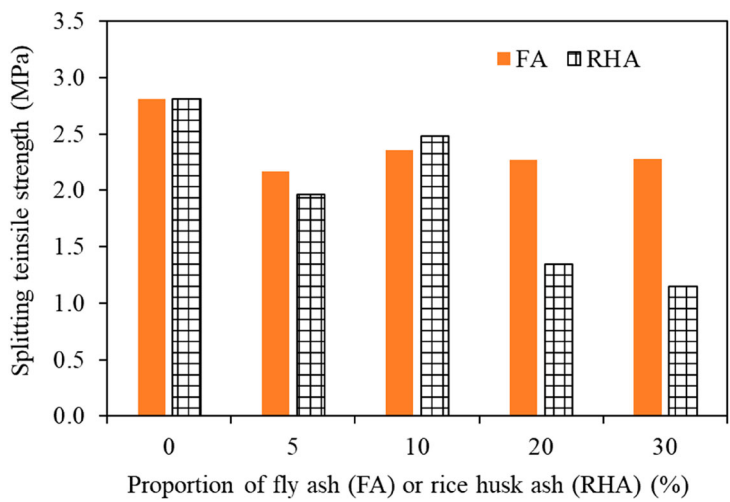


Figure 7. The 28 – day-splitting tensile strength results of RCC mixtures containing fly ash (FA) or rice husk ash (RHA).

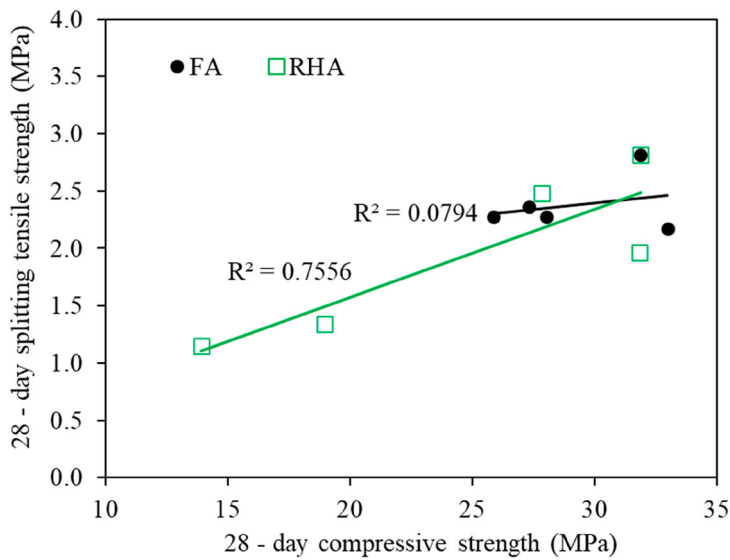


Figure 8. Correlation between splitting tensile strength and compressive strength results of RCC mixtures.

strength. Interestingly, the tensile strength values of FA and RHA mixtures were all between 7% and 9% of their corresponding compressive strengths. In contrast, tensile strengths of normal concrete are typically 10–12% of their counterpart compressive strengths. However, the splitting tensile strength results reported in the present study were obtained from testing 100 mm cubes as per GB/T 50081 (2019), rather than standard testing of 150 dia. x 300 mm height cylinders (ASTM C496, 2004).

For normal concretes, it is known that the splitting tensile strengths obtained from cubes are 10–15% higher than those obtained from the standard cylinder tests (Zabihi, 2012). However, these conversions are not standardised and were, therefore, not implemented in the present study. Moreover, to the best knowledge of the authors, no similar conversions or correlations have been established for RCC. For this reason, no attempt was made in this study to derive a mathematical relationship between the two parameters. Further research is underway to compare the splitting tensile strengths

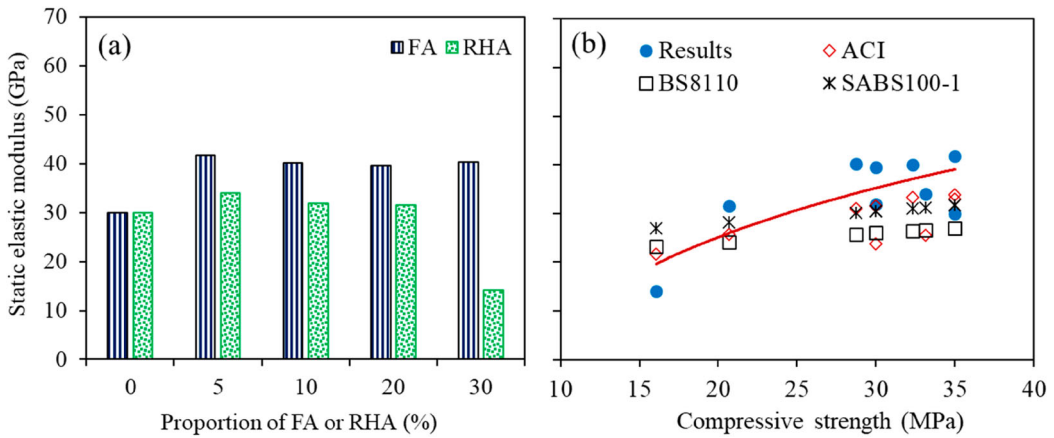


Figure 9. Static modulus of elasticity (E_c) results showing (a) effects of fly ash (FA) or rice husk ash (RHA) and (b) correlation with compressive strength.

of RCC mixtures, relating the results obtained from cubes versus those determined using the standard cylinders. For practical purposes, tensile strength is typically not employed as a design criterion. Rather, the specification of strength requirements for the design of RCC pavements is based on compressive strength and flexural strength, as earlier discussed in Section 1.1.1.

3.4. Static modulus of elasticity

It may be recalled that E_c values and compressive strengths of the RCC mixtures were determined from tests done on 150 dia x 300 mm cylinders and 100 mm cube samples, respectively (Section 2.2). The effects of FA or RHA on E_c values are shown in Figure 9(a). It can be seen that RCC mixtures containing up to 30% FA showed no E_c reduction, as similarly observed for mixtures containing up to 20% RHA. An adverse effect on E_c was observed only in the mixture containing the higher SCM content of 30% RHA. The foregoing observations confirm the findings from strength results (Sections 3.2 and 3.3), which also showed the same trends of the SCM effects.

Figure 9(b) shows the correlation between compressive strength and E_c , which gives the typical curvilinear relationship, as also reported in the literature (Rahmani et al., 2020). It is well-established that there is no single unique equation between E_c and compressive strength that applies to all concretes, as aggregates significantly influence the strength – E_c relationship. However, there are generalised standard strength – E_c models recommended for estimating E_c for use in design. Equations (2)–(4) give the standard strength – E_c models for normal concretes, as recommended in ACI 318 (1999), BS 8110 (1997) and Fulton's (2001), respectively. In the present study, these standard models were evaluated to determine their potential applicability to employ for RCCs. Included in Figure 9(b) are plots of data giving the estimated E_c values calculated using the models. It can be seen that the models gave reasonable estimates of E_c values of RCC mixtures, especially at lower strengths ≤ 25 MPa. At strength values exceeding 30 MPa, however, the models seemed to underestimate the E_c values of RCC mixtures.

$$E_c = 0.043[\rho^3 f_c]^{1/2} \quad (2)$$

where ρ is the density (kg/m^3) and f_c is the cylinder strength (MPa). In the present study, cube strength was taken as $f_{cu} = 1.25 f_c$.

$$E_{c,28} = K_o + 0.2f_{c,28} \quad (3)$$

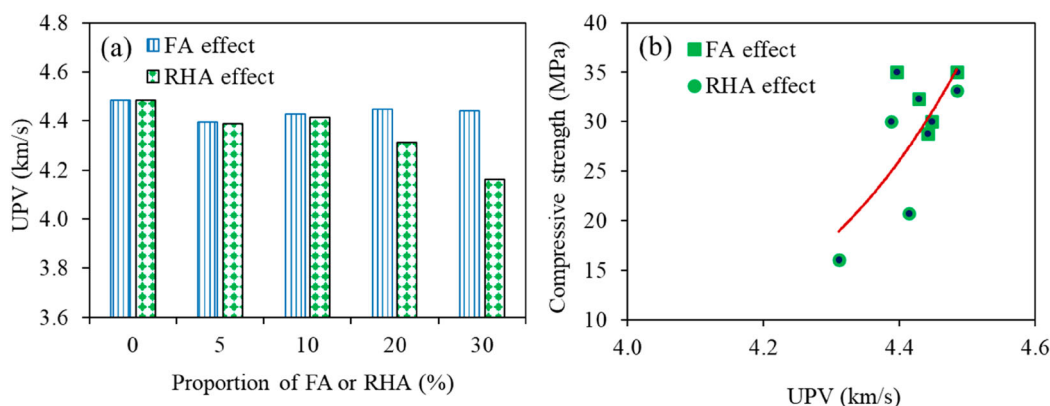


Figure 10. Ultrasonic pulse velocity (UPV) of RCC mixtures showing (a) effects of fly ash (FA) or rice husk ash (RHA) and (b) correlation between compressive strength versus UPV.

where K_o is 20 (GPa) for normal concrete and f_{c28} is the 28 – day cube strength (MPa).

$$E_{c,28} = K_o + \alpha f_{c28} \quad (4)$$

where K_o is the aggregate stiffness factor, α is the coefficient (GPa/MPa) and f_{c28} is the 28 – day cube strength (MPa). The assumed values, $K_o = 0.23$ and $\alpha = 0.25$, were selected being unique to the South African quartzite aggregate type used (Section 2.1; Fulton's, 2001).

3.5. Ultrasonic pulse velocity and dynamic modulus of elasticity

UPV measurements indicate the general quality of the interior concrete. The presence of voids, cracks and other defects or flaws that undermine concrete properties, impact UPV values which in turn capture these adverse effects to indicate overall quality. UPV values above 3000 m/s are considered to be of satisfactory quality, with higher values exceeding 3500 and above 4000 m/s indicating good to excellent quality. Values below 3000 m/s show a loss of integrity and accordingly indicate poor concrete quality (Neville, 1996).

Figure 10(a) shows the effects of FA and RHA on UPV results. The UPV value of the control was 4485 m/s, while the 5–30% FA mixtures gave UPV results that were between 4396 and 4485 m/s all of which indicate excellent quality. The 5–30% RHA mixtures also gave UPV values of 4162–4388 m/s, being of excellent quality. However, there was an interesting difference between the effects of FA and those of RHA. It can be seen in Figure 10(a) that FA mixtures showed no significant decrease in UPV values for all the mixture proportions. In contrast, the RCC mixtures containing 5–10% RHA also showed no significant UPV reductions, but at the higher proportions of 20–30% RHA, the values significantly decreased with a rise in SCM content. The foregoing observations are consistent with findings from strength results, showing that 5–10% RHA mixtures exhibited optimal performance (Section 3.2).

Figures 10(b) and 11(a) show relationships between UPV, compressive strength and dynamic moduli results based on measured data of the various RCC mixtures. Meanwhile, Figure 11(b) compares the estimated MoE determined from UPV results versus measured values. It can be seen in Figure 10(b) that UPV values increased with a rise in compressive strength, exhibiting an exponential relationship, as similarly reported in the literature (Rahmani et al., 2020). Figure 11(a) shows that the strength – E_d correlation is of the same trend as the strength – E_c relationship (Figure 9b). However, E_d values are typically higher than E_c values, an observation attributed to the absence of creep and load effects on the former. Equation (5) gives the standard relationship between E_c and E_d values of normal concretes (Fulton's, 2001; SANS, 10100-1, 2000). A plot of measured versus estimated E_c values (from Equation (5)) of RCCs, can be seen in Figure 11(b). Evidently, data points fall along the line of equality, indicating

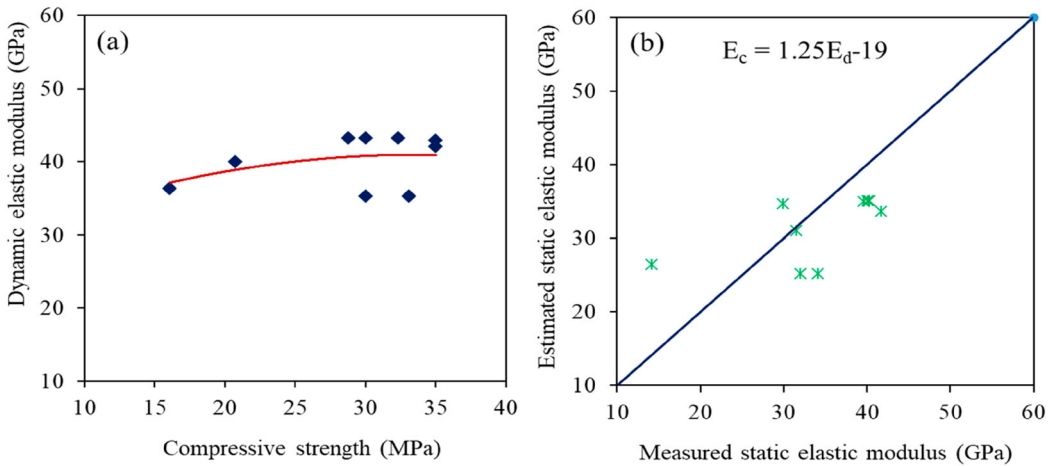


Figure 11. Relationship between (a) dynamic elastic modulus (E_d) and compressive strength and (b) measured versus estimated static elastic moduli.

that the standard Equation (5), used for normal concretes, is also applicable to RCCs.

$$E_{ce} = 1.25E_d - 19 \text{ (GPa)} \quad (5)$$

where E_{ce} is the estimated static elastic modulus and E_d is the dynamic elastic modulus.

3.6. Limitations of this study

RHA and FA effects on mechanical properties are partly related to the influence of the SCMs on ITZ. Also strongly influenced by ITZ are durability properties. This aspect is highlighted in Section 3.3, however, the present study has not focussed on the effects of the SCMs on durability properties, such as porosity, permeability and resistance to chemical attack processes. Further research is needed to fully understand the foregoing aspects related to the durability effects of RHA and FA on RCC mixtures.

4. Conclusions

An investigation was done to evaluate and compare the effects of fly ash (FA) and rice husk ash (RHA), on the mechanical properties of roller-compacted concrete (RCC) for use as pavement construction material. The two different supplementary cementitious materials (SCMs) were incorporated into RCC mixtures at proportions of 0–30% wt FA or RHA. The water content quantities used to prepare RCC mixtures were determined using the soil–compaction approach typically employed to obtain the moisture–dry density relationship. It was found that mixtures containing 5–10% wt RHA or FA gave optimum performance results, hence this range of proportions can be considered suitable for incorporation of the SCMs into RCC mixtures. Being waste by-products, several engineering and environmental benefits can result from the use of the SCMs in RCC pavements. Subsequently given are specific findings drawn from the foregoing study.

- (1) FA showed no effect on water demand, whereas RHA increased the water demand of mixtures containing higher SCM proportions comprising 20–30% wt.
- (2) Incorporation of 5–10% wt RHA into mixtures enhanced the early-age compressive strength, but showed no significant effect on late strength growth. In contrast, mixtures containing 5–10% wt FA showed early – age strength reduction, but exhibited late-age compressive

strength gain to attain values similar to or higher than those of the control, a typical tendency attributed to pozzolanic reaction.

- (3) RCC mixtures containing up to 10% FA and 10% RHA gave 28-day compressive strengths exceeding 27.3 MPa and 28 MPa, respectively, which meet the required 28–41 MPa typical values specified for RCC pavement mixtures.
- (4) The RCC mixtures containing 5–30% FA, all gave similar values of splitting tensile strength results. In contrast, RHA mixtures showed a decrease in tensile strength values with a rise in SCM proportion.
- (5) The different ACI 318, BS 8110 and SANS 10100-1 standard models for estimating static elastic modulus (E_c) from compressive strength values, were found to be valid for RCCs regardless of FA or RHA content. However, at higher strengths exceeding 30 MPa, the models seemed to underestimate the E_c values of RCCs.
- (6) The results showed that the standard equation used to estimate E_c from the dynamic modulus of elasticity (E_d) is valid for RCCs in the same manner as employed for normal concretes.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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